

1. ~~임계수=미분계수=0~~ / $f(x) = x^{\frac{2}{3}} \Rightarrow f'(x) = \frac{2}{3} x^{-\frac{1}{3}}$

$$g'(x) = \frac{1}{3} x^{-\frac{2}{3}} + \frac{2}{3} x^{-\frac{1}{3}} = \frac{1}{3} x^{-\frac{1}{3}} (x' + 2)$$

$$\therefore x = 0, -2$$

하지만 $0 \notin \text{Dom}(g(x)) \rightarrow x^{\frac{1}{3}}$ 에서 $x \neq 0$ (분모 $\neq 0$)
따라서 $x = -2$ 뿐이다

$$\therefore \text{임계수} = -2$$

2.

$$\lim_{x \rightarrow 0} \ln(1 + \sin^2 x) \cot(\ln^2(1+x)) = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin^2 x)}{\tan[\ln^2(1+x)]}$$

$$\stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow 0} \frac{\frac{1}{1+\sin^2 x} \cdot 2 \sin x \cdot \cos x}{\sec^2[\ln^2(1+x)] \cdot 2 \cdot \ln(1+x) \cdot \frac{1}{1+x}}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{\ln(1+x)}$$

$$\stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow 0} \frac{\cos x}{\frac{1}{1+x}} = 1$$

Ma Maclaurin series $\Rightarrow \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

충분히 작은 x 이라 하면,

$$\sin x \approx x \quad \ln(1+x) \approx x$$

$$\lim_{x \rightarrow 0} \ln(1 + \underbrace{\sin^2 x}_{x^2}) \cdot \cot[\ln^2(1+x)] \approx \lim_{x \rightarrow 0} x^2 \cot(x^2)$$

$$= \lim_{x \rightarrow 0} \frac{x^2}{\tan(x^2)} = \frac{1}{1} = 1$$

$$(\because \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1)$$

3.

$$\frac{1}{x} \stackrel{?}{=} \lim_{t \rightarrow 0} \left(\frac{3^t + 4^t}{2} \right)^{\frac{1}{t}} = 2\sqrt{3}$$

$$y = \left(\frac{3^t + 4^t}{2} \right)^{\frac{1}{t}} \rightarrow \ln y = \frac{1}{t} \ln \left(\frac{3^t + 4^t}{2} \right)$$

$$\lim_{t \rightarrow 0} \ln y = \lim_{t \rightarrow 0} \frac{\ln(3^t + 4^t) - \ln 2}{t} \stackrel{\frac{0}{0}}{=} \lim_{t \rightarrow 0} \frac{3^t \ln 3 + 4^t \ln 4}{3^t + 4^t}$$

$$= \frac{\ln 3 + \ln 4}{2} = \frac{\ln 12}{2}$$

$$\therefore \lim_{t \rightarrow 0} y = e^{\frac{\ln 12}{2}} = e^{\ln 12^{\frac{1}{2}}} = e^{\ln \sqrt{12}} = e^{\ln 2\sqrt{3}} = 2\sqrt{3}$$

4.

$$\sqrt[3]{990} \approx x \Leftrightarrow x^3 - 990 = 0$$

$$f(x) \quad f'(x) = 3x^2$$

$$x_1 = 10 \text{ 이다. } \text{뉴턴의 방법} \quad (x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)})$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 10 - \frac{10}{300} = \frac{299}{30}$$

$$\therefore x_2 = \frac{299}{30}$$

5.

$$f(x) \leq \tan^{-1} x$$

$$f(x) - f_0 \stackrel{\text{MVT}}{=} f'(c)(x-0), \quad c \in (0, x)$$

$$\tan^{-1} x = \frac{1}{1+c^2} \cdot x, \quad c \in (0, x)$$

$$\Rightarrow \frac{x}{1+x^2} < \tan^{-1} x < \frac{x}{1+0^2}$$

6.

$$\text{가정: } f(a) = f(b) = f(r), \quad (a < \alpha < \beta < \gamma < b)$$

$$f(a) = f(b) \text{ 이 } f: \text{ 연속인 구간 } \xrightarrow{\text{ Rolle }} f'(c_1) = 0, \quad c_1 \in (a, b)$$

$$f(b) = f(r) \quad \dots \quad \Rightarrow f'(c_2) = 0, \quad c_2 \in (b, r)$$

$$f'(c_1) = f'(c_2) \text{ 이 } f': \text{ 연속인 구간}$$

$$\xrightarrow{\text{ Rolle }} f''(c) = 0, \quad c \in (c_1, c_2) \subseteq (a, b)$$